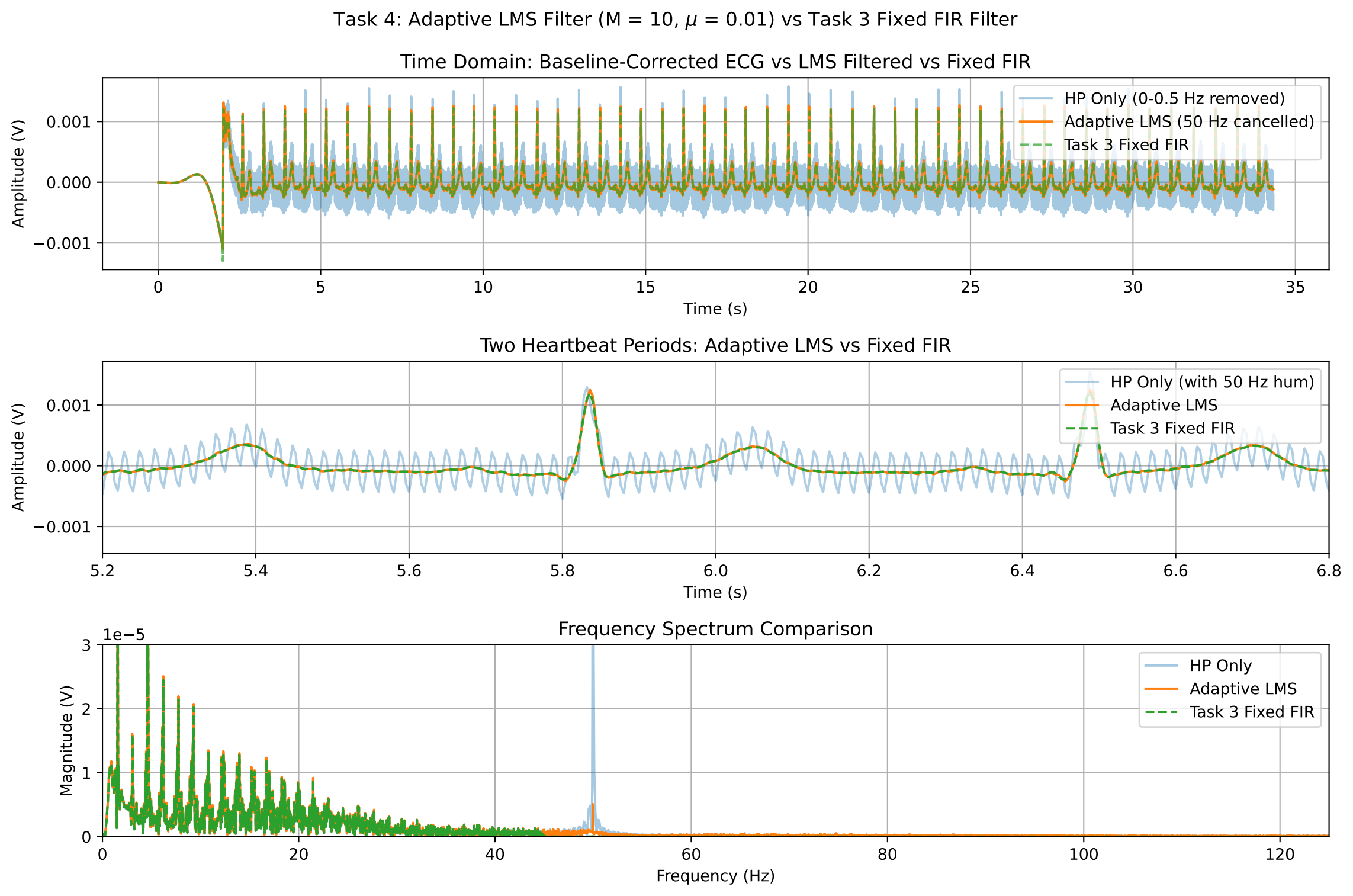

Task 4: Adaptive LMS Filter (M = 10, μ = 0.01) vs Task 3 Fixed FIR Filter
Time Domain: Baseline-Corrected ECG vs LMS Filtered vs Fixed FIR
HP Only (0-0.5 Hz removed)
Adaptive LMS (50 Hz cancelled)
Task 3 Fixed FIR
Amplitude (V)
Time (s)
Two Heartbeat Periods: Adaptive LMS vs Fixed FIR
HP Only (with 50 Hz hum)
Adaptive LMS
Task 3 Fixed FIR
Amplitude (V)
Time (s)
Frequency Spectrum Comparison
1e−5
HP Only
Adaptive LMS
Task 3 Fixed FIR
Magnitude (V)
Frequency (Hz)